Rickettsial Disease

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Introduction

Rickettsial disease is a zoonotic infection with acute febrile illness. It belongs to genus *Rickettsia* within family Rickettsiaceae which is an obligate intracellular pleomorphic gram-negative coccobacillus. The disease has been the most covert emerging and reemerging diseases and have been reported in many states and union territory in India. It is seen in both rural and urban populations.

The presenting features are nonspecific. Hence, diagnosis becomes difficult and is based on a high index of clinical suspicion. It is easily treatable when diagnosed early, but can cause significant morbidity and mortality when left untreated.

Classification

Classically rickettsial disease is divided into spotted fever and typhus group **(Table 1)**. There is still a controversy regarding the phylogeny. They are arthropod-borne diseases where humans are accidental hosts.

Classification

TABLE 1: Classification of rickettsial disease.

S. No	*Disease*	*Rickettsial agent*	*Insect vector*	*Mammalian reservoir*
i.	Typhus group			
	a. Epidemic typhus	R. prowazeki	Louse	Humans
	b. Murine typhus	R. typhi	Flea	Rodent
ii.	Scrub typhus	R. tusugamushi	Mite	Rodent
iii.	Spotted fever group*			
	a. India n tick typhus	R. conorii	Tick	Dog/rodents
	b. Rocky mountain spotted fever	R. rickettsii	Tick	Dogs/rodents
iv.	Others			
	a. Rickettsial pox	R. akari	Mite	Mice
	b. Q fever	C. burnettii	Nil	Cattle sheep goat
	c. Trench fever	Rochalimaea quintana	Louse	Humans
	d. Ehrlichioses	Ehrlichia	Tick	Deer/dog
	e. Anaplasmosis Anaplasmaphagocytophilium	Tick	Deer/dog	

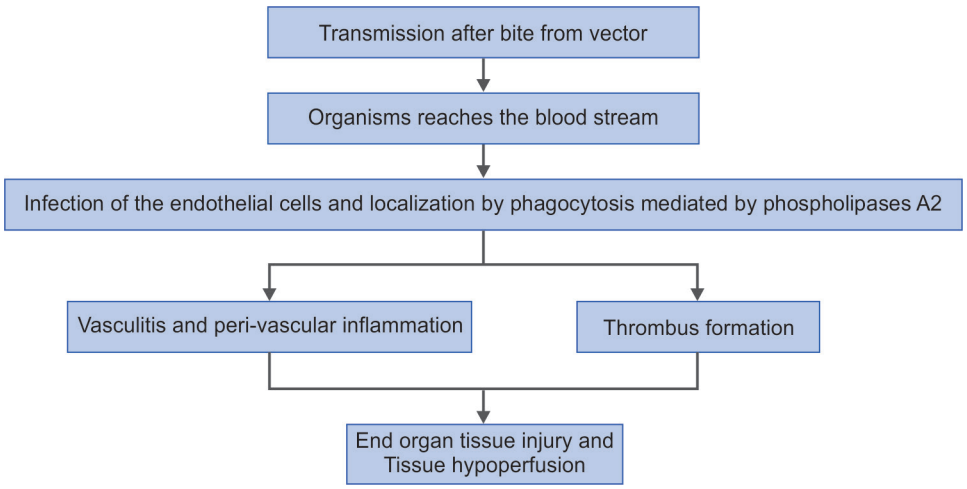
*More than 19 types of spotted fever varieties are described depending upon the geographical area where these are prevalent.

Epidemiology

India being in the part of famous Tsutsugamushi Triangle, entire India is endemic for scrub typhus. Scrub typhus is the most prevalent rickettsial infection in India which is followed by Indian tick typhus followed by murine typhus.

Pathogenesis

Flowchart 1: Pathogenesis of rickettsial disease.



Clinical Features (Box 1)

- ☑ Incubation period is 1–2 weeks
- ☑ Seasonal in many parts of India
- ☑ Eschar—pathognomonic sign of scrub typhus (seen in 7–97% cases)
- ☑ Rashes—seen in scrub typhus are transient and hence missed. Rashes are very common in Indian tick typhus, it starts as petechial rash which then progresses to palpable purpura sometimes necrosis and gangrene of digits, ear lobule, scrotum, and rarely entire limb.

BOX 1: Compatible clinical scenarios for rickettsial infection.

One or more of the following:

- ☑ Undifferentiated fever for >5 days
- ☑ Sepsis of unclear etiology
- ☑ Fever with rash or eschar
- ☑ Fever with edema
- ☑ Dengue-like disease
- ☑ Fever with headache and myalgia
- ☑ Fever with hepatosplenomegaly and/or lymphadenopathy
- ☑ Aseptic meningitis/meningoencephalitis/acute encephalitis syndrome
- ☑ Fever with cough and pulmonary infiltrates or community-acquired pneumonia
- ☑ Fever with acute kidney injury
- ☑ Fever with acute gastrointestinal or hepatic involvement

Clinical Features (Box 2)

BOX 2: Suggested epidemiological features for rickettsial infections.

- ☑ Tick bite or tick seen on clothes or surrounding areas
- ☑ Visit to common habitats of vectors—grassy or shrubby fields
- ☑ Animal sheds in proximity of home
- ☑ Contact with infected pets or stray dogs
- ☑ Living in or travelling to endemic areas
- ☑ Occurrence of similar cases in the region
- ☑ Exposure to rodents

Laboratory Diagnosis

TABLE 2: Laboratory diagnosis.

Laboratory findings	☑ Raised liver enzymes (transaminases) ☑ Thrombocytopenia ☑ Leukopenia or normal white blood cell (WBC) ☑ Neutropenia ☑ Hyponatremia ☑ Hypoalbuminemia
Serology test	☑ Positive after 1 week of illness ☑ Fourfold rise in two serum samples, 2–4 weeks apart ☑ Weil-Felix: Single titer > 1:80 suggest possible infection. It has low sensitivity but high specificity for titer or 1:320 ☑ Immunoglobulin M (IgM) enzyme-linked immunosorbent assay (ELISA)—high sensitivity and high specificity ☑ Immunofluorescence antibody (IFA)—gold standard
Polymerase chain reaction (PCR)	☑ In first week of diagnosis ☑ Done on whole blood, eschar or skin biopsy, and eschar scrapings
Western blot and indirect immunoperoxidase assay (IPA)	Not routinely available
Imaging	Chest X-ray shows infiltrates, mostly bilateral

Case Definitions

TABLE 3		
1	Suspected/ clinical case	Acute undifferentiated febrile illness of 5 days or more and compatible with **Boxes 1 and 2**; in the absence of alternate diagnosis.
2	Probable case	Suspected case with/without eschar or having rapid (<48 hours) defervescence with anti-rickettsial therapy Or Having Weil-Felix test positive with titer of 1:80 or more in OX2, OX19, or OXK Or Positive IgM ELISA for rickettsia (optical density > 0.5)
3	Confirmed case	Rickettsial DNA is detected in eschar sample or whole blood by PCR Or Fourfold rise in antibody titers on acute and convalescent sera by IFA or IPA

(ELISA: enzyme-linked immunosorbent assay; IFA: immunofluorescence assay; IgM: immunoglobulin M; IPA: immunoperoxidase Assay; PCR: polymerase chain reaction)

Comparison between the Commonest Causative Organisms in India

TABLE 4		
Feature	***Indian tick typhus***	***Scrub typhus***
Causative organism	*Rickettsia conorii*	*Orientia tsutsugamushi*
Vector	Various tick genera *(Rhipicephalus, Ixodes,* and *Boophilus)*	Trombiculid mite/chiggers
Mamalian host/ reservior	Dogs/rodents	Rodents
Epidemiological risk	☑ Endemic in Maharashtra, Karnataka, Tamil Nadu, Telengana, and Kerala ☑ Woodlands ☑ Large domestic animals in vicinity	☑ Endemic India (Tsutsugamushi Triangle) ☑ Any terrain from sea levels to himalayan foot hills ☑ Woods and clearings
Rash morphology	☑ >90% have appearance on 3–4 days ☑ Begins as faint macule and becomes petechial later ☑ First appears on extremities and spread centrally ☑ Involves palm and soles	☑ 40–50% appearance on day 4–5 ☑ Macular/maculopapular ☑ Rarely petechial ☑ Starts on trunk and spreads to peripheries.
Eschar	3–5% (uncommon)	☑ 5–80% (pathognomonic) ☑ Needs to be actively searched for
Other significant findings	☑ Gangrene of digits/earlobes leading to autoamputation ☑ GI symptoms (diarrhea/abdominal pain) ☑ Peripheral edema	☑ Regional lymphadenopathy ☑ CNS involvement ☑ (Typhus = clouding of sensorium) ☑ Sensorineural hearing loss
Weil-Felix	☑ OX2 positive ☑ OX19 positive ☑ OXK negative	☑ OX2 negative ☑ OX19 negative ☑ OXK positive
Treatment of choice	Doxycycline	Doxycycline

(CNS: central nervous system; GI: gastrointestinal)

Differential Diagnosis

- ☑ *Viral diseases*: Enteroviral disease, measles, dengue fever, chikungunya, and infectious mononucleosis
- ☑ *Bacterial disease*: Meningococcemia, leptospirosis, typhoid fever, scarlet fever, secondary syphilis, and infective endocarditis
- ☑ *Protozoal diseases*: Malaria
- ☑ *Vasculitis*: Kawasaki disease and thrombotic thrombocytopenic purpura
- ☑ Adverse drug reactions, etc.

Treatment

- ☑ Start empiric treatment without waiting for laboratory confirmation
- ☑ Do not stop treatment on basis of negative test results only
- ☑ Continue concomitant treatment for other life-threatening conditions while awaiting laboratory confirmations (e.g., meningococcemia)
- ☑ *Doxycycline is the drug of choice*:
  - *Route*: Oral or intravenous
  - *Dose*: <40 kg—2.2 mg/kg twice daily
    >40 kg—100 mg twice daily
  - *Duration*: Total of 7 days or 3 days after fever subsides or 10 days in complicated or severe cases
- ☑ *Alternative drugs*:
  - Macrolides (oral clarithromycin or oral/IV azithromycin)
  - Chloramphenicol—IV at 50–100 mg/kg/day divided 6 hourly
  - Rifampicin—should not be routinely used for treatment
  - Azithromycin—dose at 10 mg/kg/day for 5 days
- ☑ If fever do not subside in 48 hours, consider alternate diagnosis or doxycycline resistant strains
- ☑ Continue management of individual complications and supportive therapy

TABLE 5: Drugs used for the treatment of rickettsial disease.

Sr. No	*Name of drug*	*Dose, route, and duration*	*Comments*
1.	Doxycycline	2.2 mg/kg/dose BD per oral or IV (maximum 200 mg) 5–7 days or for at least 3 days until the patient is afebrile	☑ Drug of choice ☑ Rapid defervesce within 48 hours ☑ IV formulation for sick patients
2.	Tetracycline	25–50 mg/kg/dose every 6 hourly per oral (maximum 2 g/day) 5–7 days or for at least 3 days until the patient is afebrile	☑ Rapid defervesce within 48 hours ☑ IV formulation for sick patients
3.	Chloramphenicol	50–100 mg/kg/day every 6 hourly (maximum 3 g/day) 5–7 days or for at least 3 days until the patient is afebrile	☑ Most common alternative for tetracycline ☑ Most common adverse effect is agranulocytosis
4.	Azithromycin	10 mg/kg/day once daily (maximum 500 mg) 5–7 days or for at least 3 days until the patient is afebrile	☑ Preferred drug in pregnancy ☑ Recommended when doxycycline resistance is present
5.	Clarithromycin	15 mg/kg/day BD 5–7 days or for at least 3 days until the patient is afebrile	
6.	Rifampicin	10 mg/kg (maximum is 300 mg) 5–7 days or for at least 3 days until the patient is afebrile	☑ Doxycycline resistance cases
7.	Fluoroquinolones	Not recommended in pediatric age group	

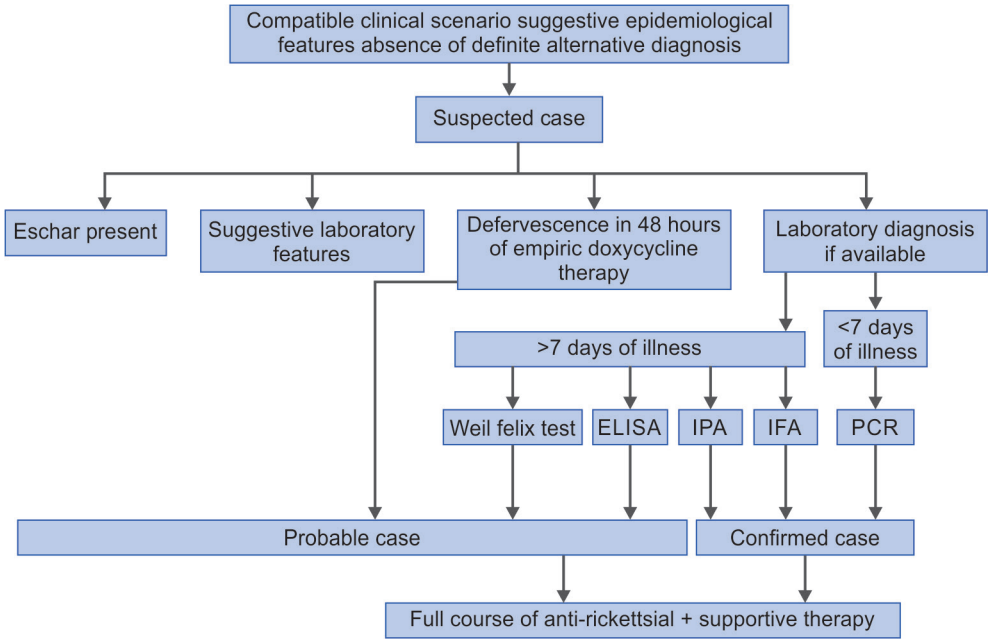
Complications

TABLE 6: Complications due to Rickettsial diseases.

System	*Complication*
Respiratory system	☑ Acute respiratory distress syndrome ☑ Pneumonitis ☑ Pulmonary edema
Cardiovascular system	☑ Myocarditis ☑ Toxic shock-like illness ☑ Congestive heart failure ☑ Pericarditis
Central nervous system	☑ Meningoencephalitis ☑ Sensorineural hearing loss
Liver and gastrointestinal (GI) system	☑ Hepatitis and liver failure ☑ Oral ulcer and GI bleed
Renal system	☑ Renal failure ☑ Hemolytic uremic-like syndrome
Skin	☑ Purpura fulminans ☑ Gangrene
Hematology	☑ Disseminated intravascular coagulation ☑ Hemophagocytic lymphohistiocytosis
Musculoskeletal	Rhabdomyolysis

Diagnosis and Management Algorithm of Rickettsial Infections

Flowchart 2



ELISA: Enzyme-linked immunosorbent assay; IPA: immunoperoxidase assay; IFA: Immunoflorescent assay; PCR: Polymerase chain reaction.

Further Reading

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